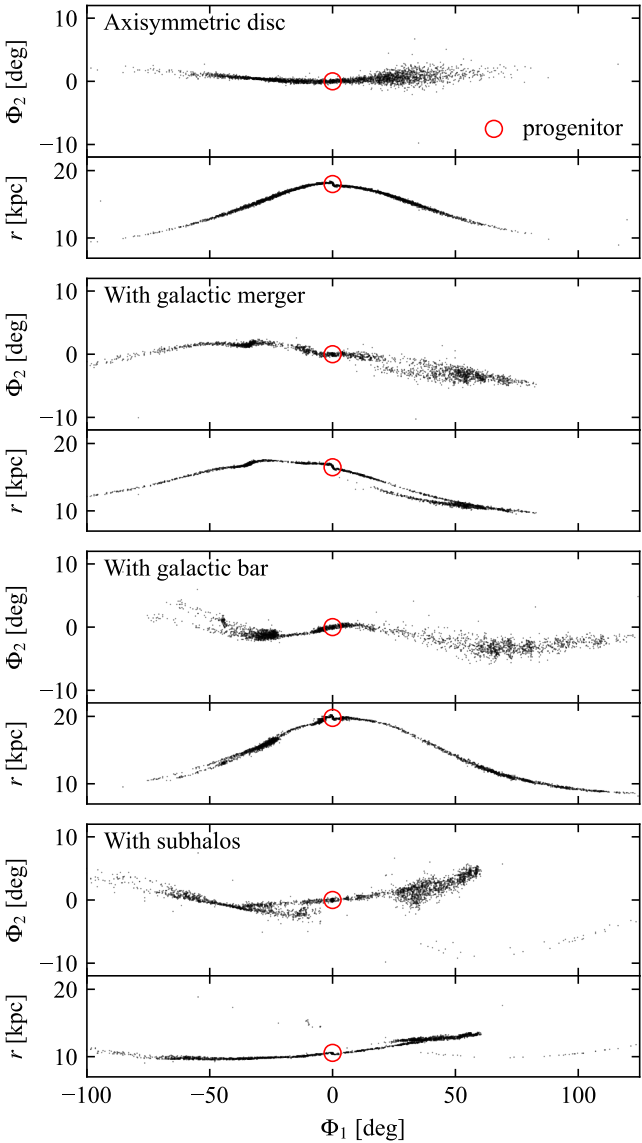

Axisymmetric disc
progenitor
With galactic merger
With galactic bar
With subhalos
Φ_2 [deg]
r [kpc]
Φ_1 [deg]
10
0
−10
20
−100
−50
50
100